Fiscal Year:	FY 2006	**Task Last Updated:**	FY 05/29/2008
PI Name:	Ferrell, Richard A. Ph.D.		
Project Title:	Theory of Phase Transitions and Simulations in Superfluid Helium		
Division Name:	Physical Sciences		
Program/Discipline:	FUNDAMENTAL PHYSICS		
Program/Discipline--Element/Subdiscipline:	FUNDAMENTAL PHYSICS--Low temperature condensed matter physics		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
PI Email:	rferrell@umail.umd.edu	**Fax:**	FY 301-314-9465
PI Organization Type:	UNIVERSITY	**Phone:**	301-405-6148
Organization Name:	University of Maryland		
PI Address 1:	2323 Physics Bldg		
PI Address 2:			
PI Web Page:			
City:	College Park	**State:**	MD
Zip Code:	20742-4111	**Congressional District:**	5
Comments:	9/27/06--PI is deceased as of 11/14/2005 per N. Hall.		
Project Type:	Ground	**Solicitation / Funding Source:**	00-HEDS-02
Start Date:	08/01/2001	**End Date:**	07/31/2006
No. of Post Docs:	0	**No. of PhD Degrees:**	
No. of PhD Candidates:	0	**No. of Master' Degrees:**	
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:	0	**Monitoring Center:**	NASA JPL
Contact Monitor:		**Contact Phone:**	
Contact Email:			
Flight Program:			
Flight Assignment:			
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Bhattacharjee, Jayanta (Indian Association for the Cultivation of Science)		
Grant/Contract No.:	NAG 3 2941		
Performance Goal No.:	1.24		
Performance Goal Text:	To explore the frontiers of physics and unlock mysteries of the universe obscured by Earth's gravity		
	1. Critical Viscosity Exponent A self-consistent mode coupling calculation of the critical viscosity exponent for classical fluids is performed and extended by the inclusion of some higher loop contributions. There are 18 different Feynman graphs corresponding to the graphs hacking 3 loops. These need to be combined into a smaller number of multiple integrals which then have to be evaluated. 2. Specific Heat in Confining Geometry a) This is the subject of CheX. It poses a serious challenge to the theory, particularly in the passage through the lambda point, where the onset of 2-dimensionality is expected. b) Fixing the Scale. The theory of confinement is compared with the earth-based data of the CHEX findings in our publication in the Journal of Low Temperature Physics (see bibliography). Although this verification of the theory is entirely satisfactory, it is incomplete in the employment of a		

Task Description:	free parameter, the geometrical scale factor for the order parameter fluctuations. Our current theoretical work is being devoted to using the idea of light scattering as a means of independently fixing the scale. To carry this out, we are finding that it may be necessary to extend the light scattering to the experimentally very difficult infared region. 3. Thermal Conductivity There is a strong incentive to move out of the quasiscaling region and closer to the lambda point and the true asymptotic region by a flight experiment (DYNAMX, R. Duncan, P. I.). In BEST (G. Ahlers, P. I.) attention is given to the effect of confinement on the thermal conductivity. 4. Superconductivity and Cosmological Analogies Because the universe is thought to have undergone some phase transitions in its earlier stages of evolutions, it is useful to compare these phase transitions with those that are familiar in condensed matter physics. The task objectives are to develop detailed theories for the flight experiments CVX (critical viscosity experiment), and CheX, and to provide theoretical background and support for the manifested low temperature flight experiments BEST and DYNAMX, as well as for other proposed experiments. The task objectives can be grouped into four categories as follows: 1. Critical viscosity exponent 2. Specific heat in confining geometry 3. Thermal conductivity of superfluid helium in confining geometry 4. Superconductivity and cosmological analogies 1. Critical Viscosity Exponent A self-consistent mode coupling calculation of the critical viscosity exponent for classical fluids is performed, including the memory effect and the vertex corrections. The incorporation of the memory effect is through a self-consistency procedure that evaluates the order parameter and shear momentum relaxation rates at non-zero frequencies, thereby taking their frequency dependence into account. This approach offers considerable simplification and efficiency in the calculation. The vertex corrections are also demonstrated to have significant effects on the numerical values for the critical viscosity exponent, in contrast to some previous theoretical work, which indicated that the vertex corrections tend to cancel out from the final result. By carrying out all of the integrations analytically, we have succeeded in tracing the origin of this discrepancy to an error in the earlier work. 2. Specific Heat in Confining Geometry It is particularly important to extend the theory below the lambda point. 3. Thermal Conductivity The good agreement between theory and the ground-based measurements by Kahn and Ahlers of the effect of confinement on the thermal conductivity indicates that it will be a reliable predictor for the ISS experiment BEST. 4. Superconductivity and Cosmological Analogies We have explored the idea that the enormous amount of energy released in a gamma ray burst might come from the crossing of two cosmic strings. The laboratory analogy of two crossed vortices in superfluid Helium-four is being studied with the goal of developing some intuitive feeling for this kind of process.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	
Task Progress:	NOTE that PI is deceased as of 11/14/2005. Grant closed out in 2006.
Bibliography Type:	Description: (Last Updated: 04/30/2008)